We have learnt to solve problems involving two sets using the formula $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. Suppose we have three sets, we can apply this formula to get a similar formula for three sets.

For any three finite sets A, B and C

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

Note

Let us consider the following results which will be useful in solving problems using Venn diagram. Let three sets A, B and C represent the students. From the Venn diagram,

Number of students in only set $A = a$, only set $B = b$, only set $C = c$.

- Total number of students in only one set $= (a + b + c)$
- Total number of students in only two sets $= (x + y + z)$
- Number of students exactly in three sets $= r$
- Total number of students in atleast two sets (two or more sets) $= x + y + z + r$
- Total number of students in 3 sets $= (a + b + c + x + y + z + r)$

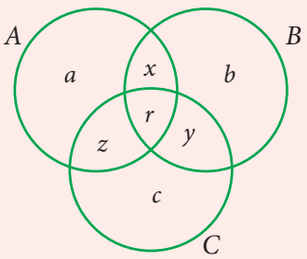


Fig.1.36

Example 1.32

In a college, 240 students play cricket, 180 students play football, 164 students play hockey, 42 play both cricket and football, 38 play both football and hockey, 40 play both cricket and hockey and 16 play all the three games. If each student participate in atleast one game, then find (i) the number of students in the college (ii) the number of students who play only one game.

Solution Let C, F and H represent sets of students who play Cricket, Football and Hockey respectively.

Then, $n(C) = 240$, $n(F) = 180$, $n(H) = 164$, $n(C \cap F) = 42$, $n(F \cap H) = 38$, $n(C \cap H) = 40$, $n(C \cap F \cap H) = 16$.

Let us represent the given data in a Venn diagram.

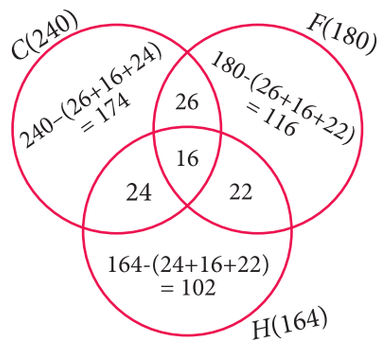


Fig.1.37

(i) The number of students in the college

$= 174+26+116+22+102+24+16 = 480$

(ii) The number of students who play only one game

$= 174+116+102 = 392$

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